Round-1

ADF & Databricks Interview

1. Explain about yourself along with the project architecture, roles and responsibilities?

What they asked me
Explain about yourself and your current project architecture.

What I said
I am working as a Data Engineer with hands-on experience in Azure Data Factory and Databricks.
In my current project, we ingest data from multiple sources like SQL Server, REST APIs, and Blob Storage using ADF. ADF pipelines orchestrate the data movement and trigger Databricks notebooks.

The raw data lands in ADLS Gen2 (Bronze layer). Databricks processes this data using PySpark, applies cleansing and transformations, and stores refined data in Silver and Gold layers using Delta Lake. The final curated data is consumed by Power BI and downstream applications.

My responsibilities include:

- Designing ADF pipelines and triggers
- Developing PySpark transformations
- Implementing Delta Lake optimizations
- Handling performance issues and failures
- Monitoring pipelines and clusters
- Coordinating with business and QA teams

Tips
Always explain:

- End-to-end flow
- Tools used at each stage
- Your personal contribution
  Interviewers mainly check ownership and clarity.

2. What is Databricks? What is the purpose of using it?

What they asked me
What is Databricks and why do you use it?

What I said
Databricks is a unified data analytics platform built on Apache Spark. It is used for large-scale data processing, analytics, and machine learning.

We use Databricks mainly because:

- It supports distributed processing using Spark
- It integrates well with Azure services
- It provides optimized execution using Delta Lake
- It supports batch, streaming, and ML workloads

Tips
Mention **performance, scalability, and Spark optimization**.
Avoid generic textbook definitions.

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3. Have you worked on PySpark? What components did you work on?

What they asked me
Which PySpark components have you worked on?

What I said
Yes, I have worked extensively with PySpark.
I have used:

- DataFrames and Spark SQL
- Transformations and actions
- Joins, aggregations, window functions
- UDFs (only when required)
- Delta Lake operations
- Performance tuning techniques

Tips
Always mention **DataFrames over RDDs** and **real transformations**.

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4. Explain Spark architecture?

What they asked me
Explain Spark architecture.

What I said
Spark architecture consists of:

- Driver: Controls execution and creates SparkContext
- Cluster Manager: Manages resources
- Executors: Execute tasks and store data in memory
- Tasks: Smallest execution units

The driver divides the job into stages and tasks, and executors process them in parallel.

Tips
Use simple flow explanation. Avoid too many internals unless asked.

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5. What is data skewness? How did you handle it?

What they asked me
What is data skewness and how did you handle it?

What I said
Data skewness happens when one partition has more data than others, causing performance issues.

In my project, I handled skewness by:

- Using salting technique
- Repartitioning data
- Using broadcast joins for small tables
- Avoiding skewed join keys

Tips
Always explain **real fix**, not just definition.

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6. Difference between Partition and Bucketing?

What they asked me
Difference between partition and bucketing?

What I said
Partition splits data physically based on column values into directories.
Bucketing distributes data into fixed number of buckets using hash function.

We used partitioning for date-based filtering and bucketing for join optimization.

Tips
Mention **where you used it in project**.

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7. What is optimization? Explain Z-ordering?

What they asked me
What optimization techniques did you use? Explain Z-ordering.

What I said
Optimization improves query performance and reduces cost.

Z-ordering reorganizes data in Delta tables so that related data is stored together. It improves data skipping for queries with multiple filter conditions.

Tips
Tie optimization with **query performance improvement**.

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8. How many data types did you work with in Databricks?

What they asked me
What data types did you work on?

What I said
I worked with:

- String, Integer, Long
- Double, Decimal
- Boolean
- Date and Timestamp
- Array and Struct

Tips
Mention **complex data types** to show experience.

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9. What is the data type of Databricks?

What they asked me
What data types are supported in Databricks?

What I said
Databricks follows Spark SQL data types such as primitive, complex, and nested types.

Tips
Keep this answer short and confident.

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10. Difference between Cache and Persistence?

What they asked me
Difference between cache and persistence?

What I said
Cache stores data only in memory.
Persistence allows storing data in memory and disk.

I used persistence when datasets were large and could not fit into memory.

Tips
Mention **why you chose persistence**.

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11. How many types of clusters? Why can't we schedule jobs on interactive cluster?

What they asked me
Explain cluster types.

What I said
There are two types:

- Interactive cluster
- Job cluster

Interactive clusters are for development and exploration. Job clusters are created temporarily for scheduled jobs, so they are cost-effective and stable.

Tips
Mention **cost and isolation**.

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12. What is the use of interactive cluster?

What they asked me
Why do we use interactive cluster?

What I said
Interactive clusters are mainly used for:

- Development
- Debugging
- Ad-hoc analysis

Tips
Do not say “for everything”.

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13. Explain medallion architecture?

What they asked me
Explain medallion architecture.

What I said
Medallion architecture has three layers:

- Bronze: Raw data
- Silver: Cleaned and transformed
- Gold: Business-ready aggregated data

It improves data quality and traceability.

Tips
Always explain **why it is used**.

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14. Have you worked on Parquet? Advantages?

What they asked me

Why Parquet?

What I said
Parquet is a columnar file format.
It provides:

- Faster query performance
- Better compression
- Efficient storage

Tips
Mention **column pruning**.

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15. What is df.explain()?

What they asked me
Have you used df.explain()?

What I said
df.explain() shows the execution plan of Spark queries.
It helps in identifying performance bottlenecks.

Tips
Mention **logical and physical plan**.

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16. How do you optimize Spark queries?

What they asked me
How do you optimize Spark queries?

What I said
I optimize by:

- Using proper partitioning
- Avoiding unnecessary shuffles
- Using broadcast joins
- Leveraging Delta optimizations
- Using Spark UI for analysis

Tips
Always connect to **real performance issues**.

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17. Tools to monitor CPU and memory?

What they asked me
How do you monitor resource usage?

What I said
I use:

- Spark UI
- Databricks Ganglia metrics
- Azure Monitor

Tips
Monitoring is frequently asked in senior interviews.

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18. Have you worked on web grid tool?

What they asked me
Have you worked on web grid tool?

What I said
I am aware of web grid tools used for monitoring and reporting, though my primary focus was Databricks and Azure monitoring tools.

Tips
Be honest, but confident.

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19. How do you upskill in AI, GenAI, LLM?

What they asked me
How do you upskill yourself?

What I said
I continuously learn through:

- Online courses
- Reading documentation
- Hands-on experiments
- Following industry updates

Tips
Show learning mindset.

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20. Are you ready to work on AI technologies?

What they asked me
Are you open to AI projects?

What I said
Yes, I am open and actively preparing myself for AI-related opportunities.

Tips
Never say no.

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21. Why do you want to join TCS?

What they asked me
Why TCS?

What I said
TCS provides long-term projects, learning opportunities, and exposure to large-scale enterprise systems.

Tips
Align with company values.

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22. What happens if your team stops working? Business impact?

What they asked me
What is the business impact?

What I said
If the team stops working, data pipelines fail, reports become outdated, and business decisions get impacted, leading to financial and operational risks.

Tips
Think from **business perspective**, not technical only.